A comprehensive analysis of the least-cost healthy diets across China from 2012 to 2023

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A comprehensive analysis of the least-cost healthy diets across China from 2012 to 2023

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Abstract

In China, shifting dietary patterns have coincided with persistent nutrient deficiencies and rising overweight and obesity, underscoring the need to promote healthy diets. This study evaluates the cost and affordability of diets aligned with four food-based dietary guidelines—the 2022 and 2016 Chinese Dietary Guidelines, the Healthy Diet Basket, and the EAT-Lancet reference diet—using nationally representative retail price data from 2012 to 2023, and assesses the nutrient adequacy and environmental implications of these least-cost diets. Nominal diet costs increased over time but grew more slowly than the food consumer price index, and real costs remained relatively stable. Affordability improved with economic growth but remains uneven across provinces and between urban and rural areas. Diets following the 2022 Chinese Dietary Guidelines, which include higher dairy recommendations, tend to be less affordable than other benchmarks. Nutrient adequacy can largely be achieved, with comparable environmental footprints. These findings highlight the importance of considering economic, nutritional, and environmental factors in dietary guideline design, and of using standardized approaches to ensure consistent and policy-relevant measurement of diet costs and affordability.

Introduction

Over the past few decades, China has experienced rapid economic growth, urbanization, and lifestyle shifts. During this period, the prevalence of undernutrition has generally declined. However, nutrient deficiencies persist, particularly in rural and low-income areas, while overweight, obesity, and diet-related chronic diseases have become increasingly prevalent. These challenges are largely driven by suboptimal dietary patterns, making the promotion of healthy diets a critical public health priority[1–3].

Food-Based Dietary Guidelines (FBDGs) serve as a foundation for food and nutrition policies, health and agricultural strategies, and nutrition education programs to promote healthy eating habits and lifestyles. They provide advice on food choices, food groups, and dietary patterns to ensure adequate nutrient intake, support overall health, and prevent chronic diseases[4]. In China, the Chinese Nutrition Society, under the leadership of multiple government agencies, periodically updates the Chinese Dietary Guidelines (CDGs) to promote balanced eating[5,6]. Meanwhile, international frameworks offer additional perspectives. The Healthy Diet Basket (HDB)[7,8], representing an average of globally representative FBDGs, serves as the reference in tracking the cost and affordability of a healthy diet[9], and the EAT-Lancet reference diet (ELD)[10,11] is considered the global reference diet promoting healthy and sustainable diets.

Since 2020, the Food and Agriculture Organization (FAO) and the World Bank have been leading global efforts to track the cost and affordability of a healthy diet (CoAHD) as a key food security and nutrition indicator[12]. This indicator provides a standardized measure to assess whether populations can afford diets following FBDGs. The CoAHD indicator is now regularly reported in flagship publication of the United Nations of the State of Food Security and Nutrition in the World (SOFI) report, offering critical insights into the economic accessibility of healthy diets across countries and regions[13]. Meanwhile, other countries have begun tracking the cost and affordability of diets based on national data systems and FBDGs[14–16].

However, while these efforts provide valuable global, regional and national insights, China has yet to systematically adopt this approach to tracking the cost and affordability of a healthy diet. Existing evidence remains limited and conceptually and methodologically fragmented, and lacks the granularity needed for detailed subnational assessments, given the stark regional and temporal variations in food prices and purchasing power over the past decade. Moreover, a comprehensive assessment of these recommended diets from the perspectives of nutrient adequacy and environmental impacts is highly needed.

To address this gap, this study first calculates the nominal and real costs of healthy diets in

China following four sets of healthy diet criteria: the CDG-2022, the CDG-2016, the HDB[8] and the updated version of the ELD[11], by determining the food items identified as least-cost per calorie in each recommended food group, using food retail price data and provincial consumer price index (CPI) data. Trends in the cost of healthy diets with the food CPI, a common measure of food inflation, are compared. Given the substantial income disparities across regions and between urban and rural areas in China, we also compare the four diet costs to daily food expenditures per capita to investigate how both cost and affordability have evolved over time across China. Finally, this study evaluates the nutrient adequacy and environmental impacts of these least-cost diets to provide a holistic perspective on affordability, nutrition, and sustainability for the least-cost diets across FBDGs.

Results

The cost of most affordable healthy diets

The HDB and ELD define food recommendations based on energy intakes, whereas CDGs use weight-based recommendations. To enable a comparative analysis, we standardized the caloric content of all four FBDGs to 2,330 kilocalories per day (kcal/d), following the HDB and global diet cost estimating calculations[7,8]. **Table S1** provides a detailed breakdown of the energy contributions from various food groups across FBDGs. Notably, the ELD features the most granular food classifications, totaling 12 distinct food groups. In contrast, CDG-2022 categorizes foods into 10 groups and CDG-2016 into 9 groups, while the HDB includes 6 groups.

Compared to its 2016 iteration, the least-cost selection of food items following CDG-2022 increases the recommended energy intake of dairy from 179 to 265 kcal, while reducing the intake of cereals from 1,143 to 982 kcal. Cereals remain the largest energy source across all FBDGs, each allocating over 32% of total energy (all above 750 kcal), with HDB assigning the highest share at 49.8% (1,160 kcal). Regarding animal-source foods including dairy, CDG-2022 recommends the highest energy contribution at 514 kcal, whereas HDB and ELD recommend more modest intakes of 300 kcal and 286 kcal, respectively. In terms of fruits and vegetables, CDG-2022, CDG-2016, and HDB emphasize relatively higher intakes at 320 kcal, 286 kcal, and 270 kcal, respectively, whereas the latest ELD suggests a slightly lower intake of 233 kcal. Notably, ELD recommends the highest energy intake from legumes, nuts and seeds (536 kcal), in contrast to the intakes of 145 kcal and 166 kcal recommended by CDG-2022 and CDG-2016, respectively. This discrepancy partly reflects differences in food group classification: both CDG-2022 and CDG-2016 categorize non-soy legumes as starchy staple foods, grouping them with whole-grain cereals. In comparison, the HDB recommends a moderate energy intake of 300 kcal for this food group.

Figure 1(A) presents trends in the least-cost estimates for the four diets, with the left panel showing nominal costs and the right panel showing real costs, adjusted for inflation using the provincial-level CPI. In nominal terms, the CDG-2022 diet is the most expensive one, reaching a mean cost of 15.8 Chinese Yuan (CNY) per day (IQR: 14.7-16.6) in 2023, closely followed by the ELD at 15.7 CNY (IQR: 14.5-16.6), and exceeding the CDG-2016 diet at 13.7 CNY (IQR: 12.7-14.2). In contrast, the HDB diet exhibits consistently lower costs throughout the study period, largely reflecting its relatively lower allocation to animal-source foods, with a mean cost of 11.8 CNY (IQR: 10.8-12.2) per day in 2023.

All four diets show an upward trend in nominal costs over time, punctuated by periodic spikes. The CDG-2022 and CDG-2016 diets show similar growth trajectories, with average annual growth rates of the mean diet cost of 2.52% and 2.48%, respectively. In comparison, the HDB and ELD diets exhibit higher mean-cost annual growth rates, at 2.64% and 2.71%, respectively (Table S2).

When examining the real costs deflated by provincial CPIs, the overall trends for the four diets remain relatively stable, with CDG-2022 rising by 0.69%, CDG-2016 rising by 0.66%, HDB rising by 0.81% and ELD rising by 0.88% per year (Table S2). This indicates that the financial burden of purchasing recommended least-cost healthy diets, relative to the broader market basket encompassing both food and non-food items, has increased only moderately over the observed period in China. However, real diet costs exhibit periods of temporarily faster growth and greater cross-provincial dispersion in certain years, consistent with short-term price fluctuations rather than sustained shifts in the underlying trend.

Figure 1(C) compares the cumulative growth of the least-cost diet for CDG-2022 with that of the food CPI across 31 provincial-level regions in China between 2013 and 2023, using 2012 as a reference year. The magnitude and direction of the growth differential vary over time, with several provinces exhibiting alternating periods in which diet costs grow faster or slower than the food CPI. Overall, however, food CPI growth outpaces the growth of the least-cost CDG-2022 diet after 2019 in nearly all provinces, except for Jiangxi and Beijing.

Diet-specific differences are also evident (Figure S1-S3). Least-cost diets following ELD and HDB show a somewhat larger number of province–year observations in which diet cost growth matches or exceeds food CPI growth, compared with diets based on the Chinese Dietary Guidelines. Nonetheless, across all dietary benchmarks, food CPI growth outpaces least-cost diet growth in most provinces, particularly after 2019.

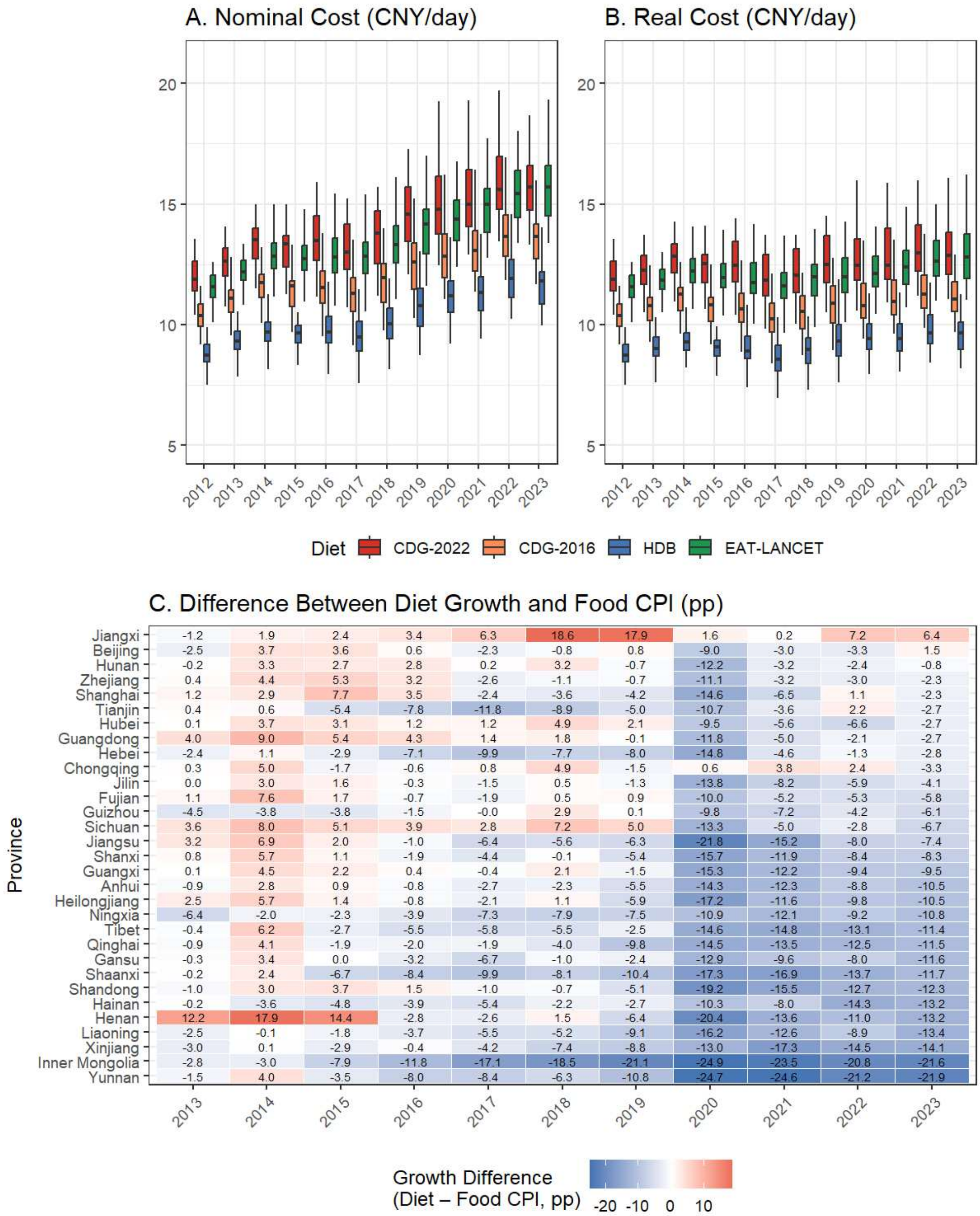

Figure 1. Least-cost of healthy diets in China from 2012 to 2023. (A) Annual distributions of the nominal least-cost estimates for four healthy diet benchmarks (CDG-2022, CDG-2016, ELD, and HDB). Diet costs are first averaged within each province and year, and boxplots summarize the distribution of provincial annual mean costs across China. Boxes represent the interquartile range (IQR), with whiskers extending to 1.5 times the IQR from the first and third quartiles, and the median indicated by the central line. (B) Same as Panel A, but for CPI-adjusted real costs. (C) Difference between the cumulative growth of the least-cost CDG-2022 diet and the cumulative growth of the provincial food consumer price index (CPI), using 2012 as the reference year. Each cell represents a province–year observation. Blue shading indicates provinces where food CPI growth exceeds healthy diet cost growth, while red shading indicates provinces where healthy diet costs have increased faster than food prices overall.

To further examine nominal cost trends across various food groups within the four healthy diet benchmarks, Figure 2 presents disaggregated costs by food group over time. Under both CDG-2022 and CDG-2016, dairy products account for a substantial share of total diet costs. Dairy costs are consistently higher under CDG-2022 than CDG-2016, making dairy the primary contributor to the higher overall cost of the CDG-2022 benchmark. Higher fruit and vegetable costs under CDG-2022 also contribute to this difference. Other animal-source foods (meat, fish, and eggs) remain at more moderate cost levels and show a noticeable increase around 2019–2020 before stabilizing or slightly declining thereafter.

Across all four benchmarks, fruits and vegetables follow broadly similar upward trajectories over the study period. In particular, fruit costs increase rapidly across all four diets, with the acceleration becoming more pronounced in the later years. In contrast, legumes, nuts, and seeds display markedly higher costs under the EAT-Lancet diet compared with the other diets, reflecting its greater emphasis on plant-based protein sources. Starchy staples provide a relatively stable baseline cost across benchmarks, while oils and fats and added sugars remain small and comparatively stable components throughout the period.

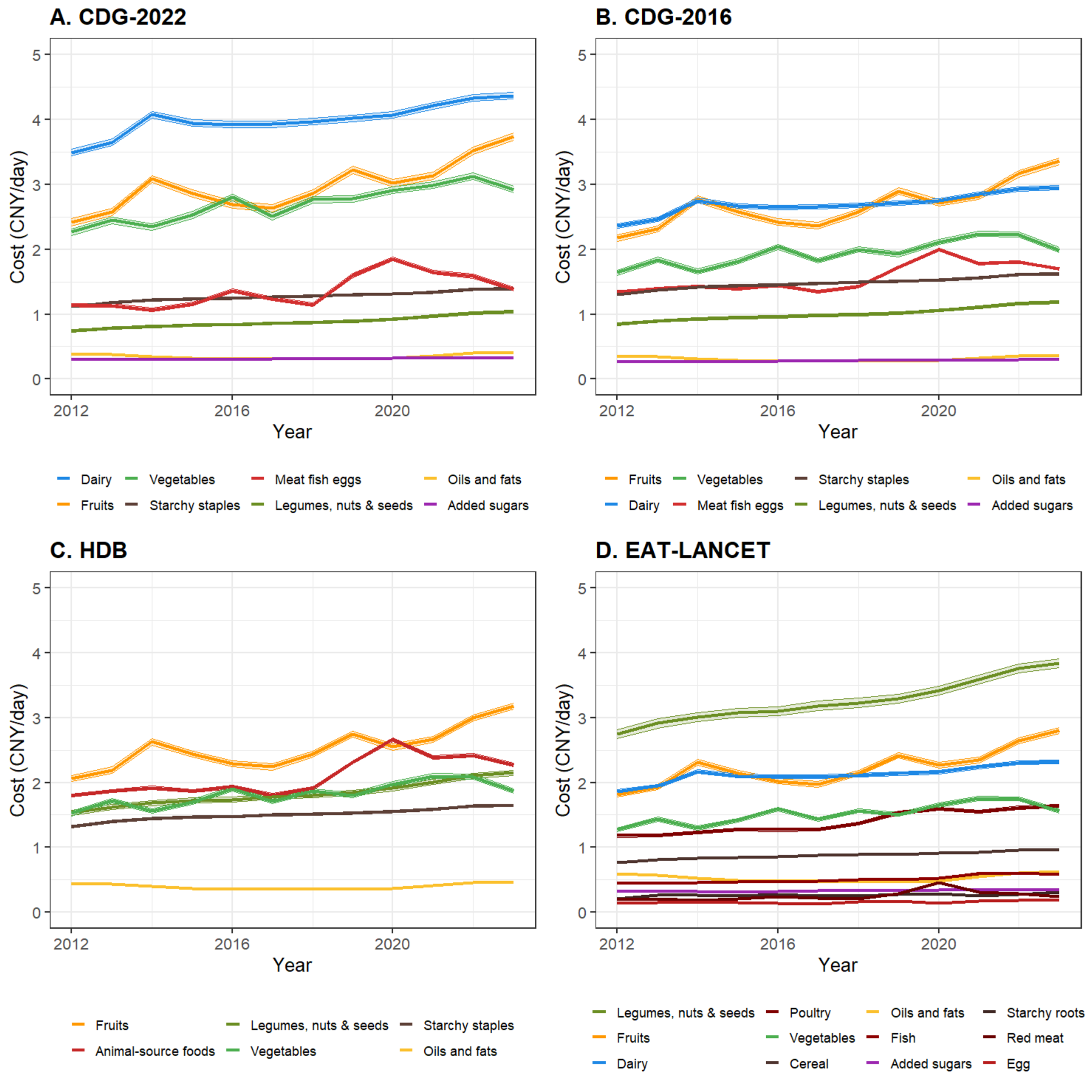

Figure 2. Trends in the least-cost of food groups within healthy diets, 2012–2023. Panels A–D show trends in the least-cost estimates for individual food groups within CDG-2022, CDG-2016, HDB, and EAT-Lancet diets, respectively. For each food group, the least cost is calculated as the average price of the least-cost foods required to meet the corresponding dietary recommendation. Annual cost trends are estimated using regression models with province and month fixed effects, which control for spatial and seasonal variations. Shaded areas denote 95% confidence intervals around the fixed-effect estimates.

Affordability of healthy diets

To assess the affordability of healthy diets, we compare the ratio between the least-cost of diet and food expenditure per capita per day [17,18]. When the ratio is below one, this indicates that, on an average basis, people can afford healthy diets using existing food expenditure without requiring additional spending. In contrast, a ratio above one implies that the least-cost of healthy diets is more than current average food expenditure, necessitating higher food spending to meet dietary recommendations. Given substantial variation in food expenditure across provinces and the disparities between urban and rural areas, Figure 3 displays this affordability ratio separately for urban and rural areas in each province between 2012 and 2023.

Figure 3 reveals a clear divergence in affordability patterns between rural and urban areas. In rural areas, the affordability of healthy diets improves substantially over time across all dietary benchmarks. For CDG-2022 and ELD diets, many provinces move closer to, and in several cases cross, the affordability threshold after 2020. Nevertheless, these two benchmarks remain the least affordable, with only about half of provinces able to afford them in 2023 based on average food expenditure per capita, led by wealthier provinces such as Shanghai, Zhejiang, Jiangsu, and Beijing. The mean affordability ratio in rural areas in 2023 was 1.02 (IQR: 0.86–1.15) for CDG-2022 and 1.02 (IQR: 0.84–1.17) for ELD.

In contrast, diets following CDG-2016 and the Healthy Diet Basket (HDB) are affordable in most rural provinces without requiring additional food expenditures on average, with mean ratios of 0.88 (IQR: 0.74–1.00) for CDG-2016 and 0.76 (IQR: 0.63–0.88) for HDB. Remaining affordability gaps are largely confined to a small number of provinces in the northwest, southwest, or northeast.

In urban areas, the least-cost of all healthy diets could be covered by existing average food expenditures in all provinces over the period, with affordability ratios consistently below one.

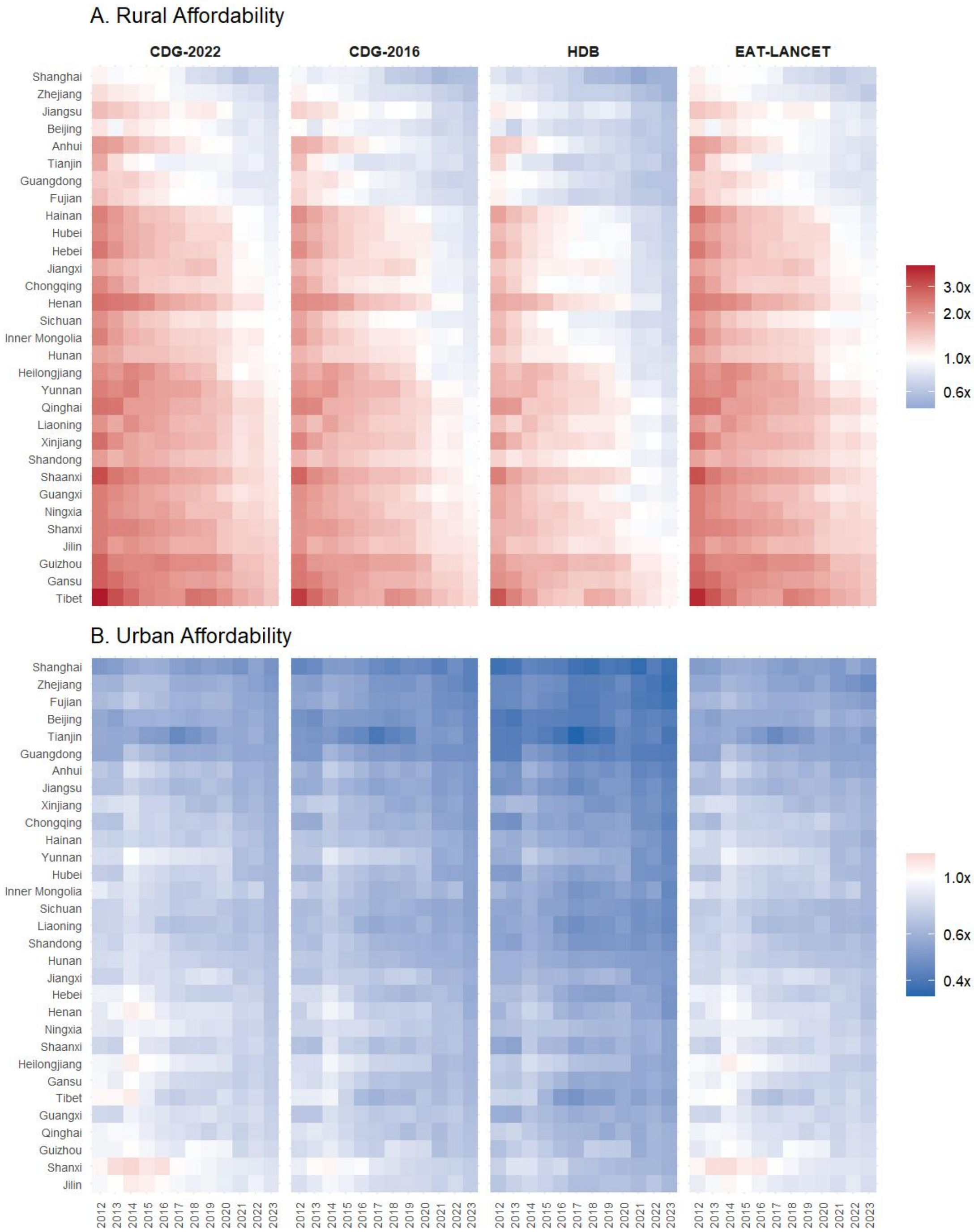

Figure 3. Provincial affordability of four healthy diet standards—CDG-2022, CDG-2016, the EAT-Lancet Diet (ELD), and the Healthy Diet Basket (HDB)—from 2012 to 2023. Each tile shows the affordability index for urban and rural populations in a given province and year, defined as the ratio of the average daily diet cost to the average daily food expenditure. Values above 1 indicate that a diet costs more than residents typically spend on food. Red tones denote lower affordability (higher relative cost), while blue tones denote greater affordability. White corresponds to an affordability index of 1, where the diet cost equals average food expenditure.

Nutrient adequacy and environmental impacts of the most affordable healthy diets

We further assess the nutritional quality and environmental impacts of the four healthy diets when constructed under a least-cost constraint. Overall, all four dietary benchmarks achieve high levels of nutritional adequacy, while differences in environmental impacts across diets remain relatively modest under low-cost settings.

Figure 4 summarizes overall nutritional adequacy and associated environmental impacts of the four least-cost healthy diets. Panel A presents the Mean Adequacy Ratio (MAR), a summary indicator of overall nutritional quality derived from Nutrient Adequacy Ratio (NAR) for protein and 12 essential micronutrients using food composition data in Chinese Food Composition Tables[19] (see methods). Across all diets, MAR values are consistently high and tightly distributed, indicating that least-cost diet construction can maintain adequate overall nutrient intake. Diets based on CDG-2022 and CDG-2016 exhibit mean MAR values of 0.96 (IQR: 0.94–0.97 and 0.94–0.98, respectively), while the ELD benchmark shows a similarly high level of adequacy (mean MAR = 0.96, IQR: 0.96–0.97). The Healthy Diet Basket (HDB) displays a slightly lower but still robust level of adequacy (mean MAR = 0.95, IQR: 0.93–0.96). The limited dispersion of MAR across diets suggests that overall nutrient adequacy is maintained under least-cost selection.

Despite high overall MAR values, nutrient-level differences remain (Table S3). For CDG-2022, most nutrients exhibit average NARs above 0.9, while selenium shows markedly lower adequacy (mean NAR = 0.68, IQR: 0.58–0.70). A similar pattern is observed for CDG-2016, with relatively lower adequacy for selenium (mean NAR = 0.81, IQR: 0.75–0.89) and niacin (mean NAR = 0.88, IQR: 0.83–0.97). In the HDB diet, riboflavin displays the lowest adequacy among assessed nutrients (mean NAR = 0.67, IQR: 0.64–0.69), accompanied by moderately lower selenium adequacy (mean NAR = 0.86, IQR: 0.81–0.95). Diets following the ELD benchmark also show lower adequacy for riboflavin (mean NAR = 0.77, IQR: 0.75–0.80) and selenium (mean NAR = 0.78, IQR: 0.76–0.81), while protein and most mineral nutrients consistently achieve near-unity NARs.

Panels B and C of Figure 4 report greenhouse gas emissions and total water use associated with least-cost diets constructed under each benchmark. Across all four dietary benchmarks, environmental footprints remain at relatively low levels, with substantial overlap in the interquartile ranges across CDG-2022, CDG-2016, and the HDB. Diets based on the ELD exhibit modestly higher greenhouse gas and water footprints. This pattern reflects the fact that tofu is the only available item used to meet the required quantities of legumes, nuts, and seeds (Table S5). As a result, tofu accounts for a substantially larger share of total dietary energy under the ELD (23%) compared with CDG-2022 (6.2%), CDG-2016 (7.1%), and the HDB (12.9%) (Table S4). Given that tofu is associated with relatively higher greenhouse gas emissions and water use per kilogram, its greater quantitative contribution under the ELD translates into moderately higher overall environmental footprints.

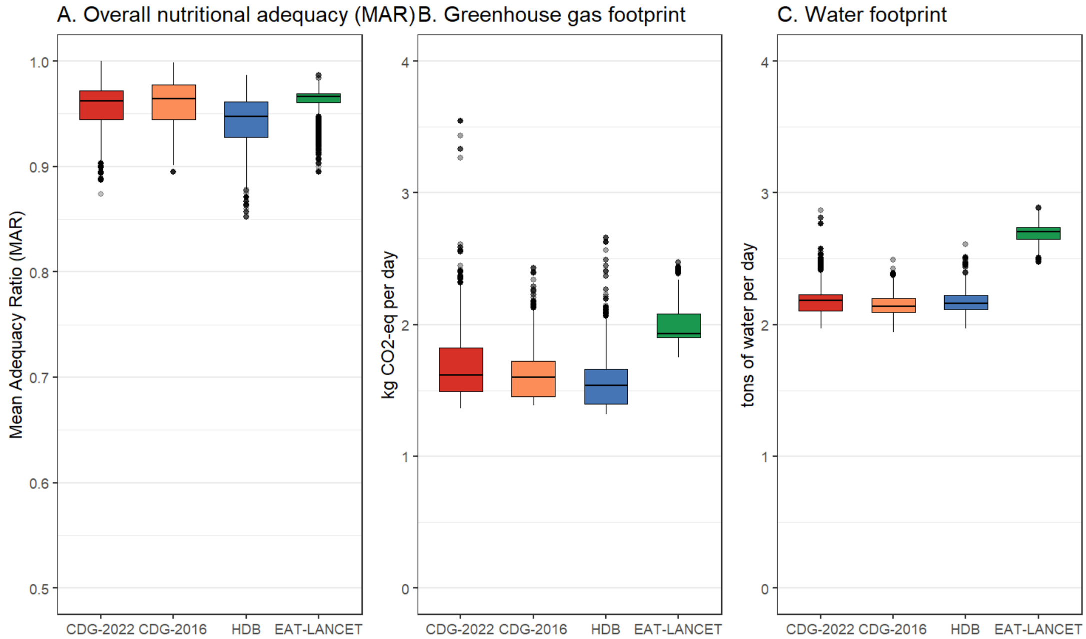

Figure 4. Nutritional adequacy and environmental impacts of least-cost healthy diets in China. Panel A shows the distribution of the Mean Adequacy Ratio (MAR), summarizing overall nutritional adequacy across protein and 12 essential micronutrients. Panels B and C present the distributions of greenhouse gas emissions (GHGE, kg CO_2-eq per diet per day) and total water footprint (tons per diet per day), respectively. Box plots represent the interquartile range (IQR), with whiskers extending to 1.5 times the IQR. The median value is indicated by the central line, and outliers are shown as individual points. All estimates are based on city–month least-cost diet solutions across provinces and years.

Discussion

In China, unhealthy dietary patterns have been increasingly linked to an increase in obesity, overweight conditions, and diet-related chronic diseases prevalence. Meanwhile undernutrition issues, including child stunting and wasting, and micronutrient deficiencies persist particularly in rural areas and low-income households. These double burdens highlight the significant nutritional challenges and emphasize the urgent need to promote healthy eating[1–3]. Ensuring that healthy diets are affordable and accessible to all is a critical prerequisite for improving dietary behaviors and public health outcomes.

Building on globally established monitoring frameworks and methodological standards for assessing the cost and affordability of healthy diets, this study applies a standardized analytical framework to the Chinese context. Using four healthy diet reference frameworks, CDG-2022, CDG-2016, the HDB, and the ELD reference diet, the analysis implements a consistent least-cost diet methodology based on nationally available food retail price data aligned with CPI food price tracking, while CPI data are used to assess real cost trends over time. The results provide a systematic and internationally comparable assessment of the cost and affordability of healthy diets in China, while also evaluating their nutrient adequacy and environmental implications. By integrating economic, nutritional, and environmental dimensions within a unified framework, this study demonstrates how globally comparable approaches for monitoring the cost and affordability of healthy diets can be implemented in large and complex national contexts such as China, based on national data systems and complemented where necessary by relevant global datasets.

Although the nominal costs of healthy diets have shown an upward trend during the evaluation period, real costs have only moderately increased. This finding suggests that China's food market has in general effectively stabilized price increases for affordable, locally available food items required by dietary guidelines. Moreover, the costs of meeting the healthy diets have generally risen at a slower pace than the food CPI in most provinces, particularly in recent years, revealing that the low-cost food items meeting requirements of healthy reference diets had lower inflation than the average foods typically consumed in China. One possible explanation is that Chinese government's practice of food stockpiling, buying food when prices are low and releasing it when prices are high, tends to involve procuring the least expensive local foods, which in turn helps stabilize the least-costs of healthy diets[20].

While the overall trend in the least-costs of healthy diets appears stable, periodic spikes have been observed. To address potential future spikes, policymakers could consider improving the early warning systems for monitoring food prices to swiftly respond to sudden increases[21,22]. Strengthening the supply chain, including improving transportation and local storage, could also help stabilize prices[23–25]. Additionally, among the diets analyzed, CDG-2022 emerges as the most expensive, exceeding the cost of CDG-2016. This difference is primarily due to its higher dairy requirements, as well as moderately higher costs associated with fruits and vegetables. The ELD also requires relatively higher costs, largely due to greater emphasis on plant-based protein sources. In contrast, the HDB selects two lowest-cost animal-source foods and does not require dairy, making it comparatively more affordable.

Despite the rising diet costs, healthy diets have become increasingly affordable in China, largely due to rapid income growth across the nation. Affordability analysis reveals that in 2023, urban residents in all provinces and rural residents in a majority of provinces could, on average, meet the cost of HDB and CDG-2016 with current food expenditures. In contrast, CDG-2022 and ELD reference diets were affordable for urban residents but required additional spending in rural areas in most provinces. Regarding the high affordability in urban areas, it is crucial to recognize the role of additional factors influencing people's dietary choices, such as nutrition and health education, as well as regulatory measures targeting unhealthy food products, such as sugar-added beverages and ultra-processed foods[26,27]. However, these findings represent overall averages, and the persistent issue of unaffordability among low-income populations, particularly in less affluent provinces and rural areas, must still be addressed. Policymakers should therefore prioritize initiatives that support vulnerable populations in accessing nutritious and affordable diets. Additionally, promoting plant-based healthy diets may also help improve affordability as well as environmental sustainability.

This study further compares the nutritional quality and environmental impacts of the four FBDGs. Our findings highlight that both the national and international dietary guidelines generally meet nutritional requirements, even when least-cost items are selected. The CDG-2022, CDG-2016 and ELD reference diets achieve particularly high levels of nutrient adequacy, while the overall mean adequacy ratio for the HDB reaches 95%, aligning with international findings[8]. However, the results also indicate that minor micronutrient gaps may exist for low-cost diets in China, particularly for selenium and certain B vitamins. One factor contributing to these results is the relatively limited food list used in this analysis. For example, only two dark-green leafy vegetables (Rape green and Chinese chives) are included in the price dataset (Table S5). Another factor is that the least-cost vegetables in China are often less nutrient-dense items with long shelf lives, such as radish and cabbage. Meeting micronutrient requirements would require people to consume more nutrient-rich plant-based food items, such as dark green leafy vegetables, and orange and red fruits and vegetables, which may increase overall diet costs as highly recommended in the CDG-2022.

Despite these potential nutritional gaps, the environmental assessment suggests that least-cost diets constructed under the four dietary benchmarks are associated with broadly comparable environmental footprints. Greenhouse gas emissions and water use remain relatively similar across CDG-2022, CDG-2016, and the HDB, with substantial overlap in their distributions, while diets based on the ELD reference diet show slightly higher environmental footprints in this analysis. This pattern is partly driven by the limited food list available for legumes, nuts, and seeds, where tofu serves as the primary substitute to meet higher recommended quantities under the ELD framework. At the same time, the relatively modest environmental footprints observed under the Chinese dietary guidelines partly reflect the structure of China's food system, where pork, rather than beef and lamb, is typically the most affordable and widely consumed animal-source food. It is also important to note that least-cost diet modelling tends to generate relatively moderate greenhouse gas emissions because lower-cost animal-source foods are typically selected[28,29]. However, red meat consumption in China has increased substantially over recent decades and is expected to continue rising with income growth and dietary transition[30]. In this context, incorporating environmental considerations into dietary guidance will become increasingly important, aligning dietary recommendations for both human and environmental well-being while maintaining affordability within the current Chinese food system[31].

In addition, the relatively high dairy recommendations in the CDG-2022 may warrant further consideration from the perspectives of affordability, cultural dietary patterns, and environmental sustainability. Dairy products are not a large component of traditional diets in China, and daily consumption, particularly the equivalent of two cups of milk per day as recommended in CDG-2022, may be difficult for many households to achieve in practice. Moreover, dairy production is associated with relatively higher greenhouse gas emission compared with many plant-based foods, which could increase both the environmental footprint and the cost of recommended diets.

This study has several limitations. First, although the food price data used capture many commonly consumed food items, such as eggs and locally available vegetables, the food list remains relatively limited. The ideal source would be CPI food price data from the National Bureau of Statistics of China for broader food coverage. Nevertheless, the retail food price data used in this study serve as a useful reference for monitoring CPI in China, and the resulting diet cost estimates are consistent with statistics reported by the World Bank and FAO, which rely on comprehensive retail food price data from the International Comparison Program[17]. Additionally, this study does not have separate urban and rural price data, which could enhance the precision of affordability estimates. However, China's food market is generally efficient, which helps keep urban-rural price gaps relatively small, reducing potential bias in the findings[32]. Second, the Chinese food composition data do not provide nutrient content for all essential micronutrients; however, the analysis based on available data aligns with international findings, supporting the reliability of the results. Lastly, the absence of representative income distribution data at the provincial level limits the ability to fully align affordability estimates with global monitoring standards[33]. The indicator applied only reflects average conditions rather than income-specific disparities, highlighting the need for future research to incorporate disaggregated income data for greater precision. Despite these limitations, this study provides comprehensive and valuable insights into the cost, affordability, nutritional quality, and environmental implications of low-cost healthy diets in China, offering a strong foundation for future research and policy discussions.

Methods

Data

Price data for 62 food items (detailed in Table S5) are obtained from CPIN[34], supported by the National Development and Reform Commission Price Monitoring Center. This extensive network includes over 5,000 sites across 31 provinces, which are directly managed by the Central Government, facilitating nationwide price monitoring. CPIN collects price data for these food items three times a month from both supermarkets and farmers' markets. This study collated mean monthly city price data for each food item from January 2012 to December 2023. In total, this study had 2,269,057 distinct city-month-food item combinations. Cost information for each representative food is determined per gram, and all cost information is presented in CNY.

Data on the consumer price indexes and average food consumption expenditure of residents in the urban and rural areas of these provinces are from the China Statistical Yearbook[32]. The expenditure covers a wide range of food-related spending, including staples, edible oils, vegetables, fruits, meats, poultry, eggs, dairy products, aquatic items, confectionery, beverages, nuts, and expenses related to dining out and takeaway.

Healthy diets

This study considered four national and international FBDGs, including CDG-2022, CDG-2016, ELD and HDB. The HDB served as the reference diet for calculating the cost of healthy diets. It was developed based on representative quantifiable national FBDGs worldwide, classifying food items into six food groups: starchy staples, vegetables, fruits, legumes / nuts / seeds, animal-source foods, as well as oils and fats. The ELD emphasizes whole grains (use cereals in the analysis), starchy roots, vegetables, fruits, nuts and legumes (combined in the analysis), dairy, poultry, fish, eggs, red meats, different oils and fats (combined in the analysis), and added sugars. The CDG-2022 categorizes foods into ten groups: starchy staples, dark-green leafy vegetables, red and orange vegetables, other vegetables, fruits, legumes / nuts / seeds, meat / fish / eggs, dairy, oils and fats, and added sugars. The CDG-2016 combined different vegetables into one group, therefore recommending foods into 9 groups. These guidelines recommend daily consumption of 11-15 different food items. The detailed dietary energy allocation and food item numbers across food group and FBDG are presented in Table S1, which is consistent with estimations in previous global study.

The least-cost and affordability

This study used CoAHD method to estimate the least-cost of the four FBDGs. The least-cost healthy diet (CoHD) was calculated by summing the cost of each food group, expressed as the product of recommended energy intake and the corresponding price per kilocalorie:

$$CoHD_{dt} = \sum_{g=1}^{G} c_{gt} \cdot E_{gd},$$

where E_{gd} is the recommended daily energy intake from food group g under dietary guideline d, and c_{gt} is the price per kilocalorie in food group g. The price per kilocalorie is defined as:

$$c_{gt} = \frac{P_{gt}}{k_g},$$

where P_{gt} denotes the retail price per edible portion of the selected least-cost food item(s) in food group g at time t, and k_g is the caloric content per edible portion. The least-cost principle was implemented by selecting the lowest-cost item(s) within each food group:

$$P_{gt} = \min_{i \in \mathcal{I}_g} p_{it},$$

where $\mathcal{I}_g$ is the set of available food items in food group g, and p_{it} is the price of item i. This formulation ensures that cost minimization is applied within each food group, while preserving the dietary structure defined by the FBDGs.

The caloric content of all four FBDGs was standardized to 2,330 kcal per day, following the HDB:

$$\sum_{g=1}^{G} E_{gd} = 2330.$$

Real costs were adjusted for inflation using the consumer price index (CPI), with 2012 as the base year:

$$RCoHD_{dt} = CoHD_{dt} / (\frac{CPI_t}{CPI_{2012}}).$$

The affordability indicator was calculated by dividing the least-cost by food expenditure per capita per day:

$$Affordability_{dpt} = \frac{CoHD_{dpt}}{FE_{pt}}.$$

where $Affordability_{dpt}$ is the affordability of dietary guideline d in province p and year t, $CoHD_{dpt}$ is the cost of the healthy diet estimated for province p in year t, and FE_{pt} denotes food expenditure per capita per day in province p in year t.

Nutritional Quality

To assess the nutritional quality of the four FBDGs, we calculate NAR and MAR. The average nutrient intake was estimated based on the least food amount recommended by the dietary guidelines using the Chinese Food Composition Tables (Standard Edition 6)[19]. Nutrient adequacy was assessed through NAR for protein and 12 essential micronutrients: calcium, iron, magnesium, phosphorus, zinc, selenium, copper, vitamin A, thiamin, riboflavin, niacin, and vitamin C. NAR was calculated as the ratio of the actual nutrient intake to its estimated average requirement (EAR), with EAR values taken from the latest Chinese Dietary Reference Intakes (DRIs) for adult females of 30-49 years old [35]:

$$NAR_{nd} = \frac{I_{nd}}{EAR_n}$$

where I_{nd} is the intake of nutrient n under dietary guideline d, and EAR_n is the estimated average requirement.

The intake of each nutrient was derived from the quantities of foods in the least-cost diet:

$$I_{nd} = \sum_{i=1}^{N} q_{id} \cdot a_{in},$$

where q_{id} is the quantity of food item $i$in diet d, and a_{in} is the nutrient content of nutrient $n$per unit of food i.

Theoretically, NAR values could be below, equal to, or above 1. However, in this study, NAR was capped at 1:

$$NAR_{nd}^{*} = \min\left(NAR_{nd}, 1\right),$$

to ensure that a nutrient with a high NAR could not compensate for a nutrient with a low NAR

The MAR, representing the overall nutritional quality of the diet, was calculated as the average of all NAR values across protein and the micronutrients：

$$MAR_d = \frac{1}{K}\sum_{n=1}^{K} NA\,R_{nd}^{*},$$

where K is the total number of nutrients considered. This provides a percentage that reflects the degree to which the diet meets established nutritional standards. An MAR value of 1.0 indicates optimal nutritional adequacy, signifying a diet that meets all recommended nutrient intakes.

Environmental impacts of diets

The environmental impacts of food consumption were assessed using the SU-EATABLE LIFE (SEL) dataset of food commodities, a publicly available multilevel dataset of carbon and water footprint values for food commodities. The SEL database compiles data from peerreviewed studies and grey literature using a standardized methodology for data screening, harmonization, aggregation, and uncertainty assessment, and provides internationally comparable estimates for food items based on harmonized functional units and system boundaries.

In this study, these food-specific footprint values were matched to the foods selected in each least-cost diet to estimate associated greenhouse gas emissions and total water use. Total environmental impacts were calculated as:

$$GHG_d = \sum_{i=1}^{N} q_{id} \cdot e_i,$$

$$Water_d = \sum_{i=1}^{N} q_{id} \cdot w_i,$$

where q_{id} is the quantity of food item i in diet d, e_i is the greenhouse gas emission factor, and w_i is the water footprint associated with food item i.

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